

Public Economics

Lecture 2 · Theoretical Tools of Public Finance

Dr. Shushanik Margaryan

Winter Term

WU Vienna University of Economics and Business

Today's Questions

- In this lecture we will review microeconomic concepts that most of you should be familiar with
- Utility function, budget constraints, indifference curves, etc.
- What is the theoretical effect of raising — or cutting — Austria's minimum-income support (**Sozialhilfe**) on economic efficiency?

2.1 Constrained Utility Maximization

2.2 Putting the Tools to Work: Sozialhilfe and Labor Supply

2.3 Equilibrium and Social Welfare

Theoretical vs. Empirical Tools

Theoretical tools

The set of tools designed to understand the *mechanics* behind economic decision making — graphical and mathematical models of utility maximization subject to constraints.

Empirical tools

The set of tools designed to *analyze data* and answer the questions raised by theoretical analysis (the subject of Lecture 3).

Theory tells us the *direction* of an effect; data tells us its *size*. We need both.

2.1 Constrained Utility Maximization

Utility function

A mathematical mapping of an individual's consumption choices into a level of well-being: $U = u(X_1, X_2, \dots)$.

- Consumers are assumed to undertake **constrained utility maximization**: maximize well-being subject to available resources.
- Key assumption: **non-satiation** (“more is better”) — more of a good is always at least as good as less.

Running example: Lena, a WU student, choosing between *coffees at the Kaffeehaus* (C) and *cinema visits* (M).

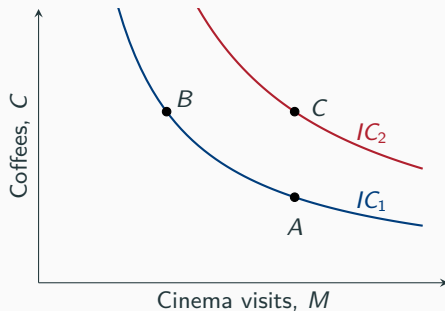
Indifference Curves

Indifference curve

All bundles of goods that leave an individual equally well off.

- Bundle *A*: 2 coffees, 1 cinema visit
- Bundle *B*: 1 coffee, 2 cinema visits
- Bundle *C*: 2 coffees, 2 cinema visits

Lena is indifferent between *A* and *B*, but strictly prefers *C* to both (more of everything).



Two properties (both from “more is better”): (1) higher indifference curves are preferred; (2) indifference curves always slope downward.

Marginal utility

The additional utility from consuming one more unit of a good: $MU_1 = \partial u / \partial X_1$.

- Example: $U = \sqrt{C \times M}$ (holding $C = 2$ fixed). Moving from 0 to 1 cinema visit raises U from 0 to $\sqrt{2} = 1.41$; from 1 to 2 visits, U rises only to $\sqrt{4} = 2$ (a gain of 0.59); from 2 to 3, the gain is only 0.45.
- This is **diminishing marginal utility**: each additional unit of a good raises utility by less than the previous unit.

Aside: Does Marginal Utility Always Diminish?

Not for addictive goods

Becker & Murphy (1988), “A Rational Theory of Addiction”: today’s utility from consuming a good can depend on *past* consumption of it — you build a habit.

- The more you have already consumed, the more you enjoy (or “need”) consuming more — the opposite of diminishing MU.
- This is a leading theoretical framework for taxing tobacco, alcohol, and sugar: if consumers systematically under-weigh future harm, corrective taxes can improve welfare beyond just correcting externalities.

Marginal Rate of Substitution

Marginal rate of substitution (MRS)

The rate at which a consumer is willing to trade one good for another; equals (minus) the slope of the indifference curve: $MRS_{C,M} = MU_M/MU_C$.

- For $U = \sqrt{C \times M}$: $MRS_{C,M} = C/M$.
- **Diminishing MRS**: as Lena has more cinema visits and fewer coffees, she is willing to give up less and less coffee for one more visit.

Diminishing MRS is exactly why indifference curves are *convex* (bowed toward the origin).

Budget constraint

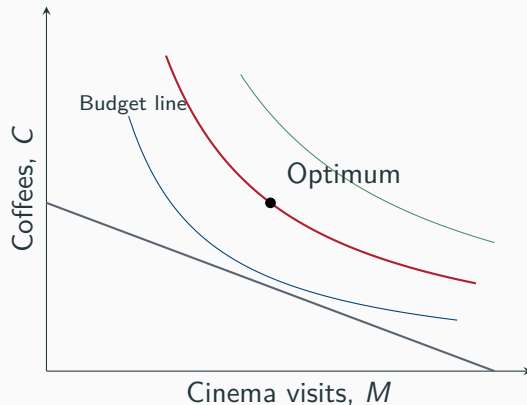
All combinations of goods an individual can afford, spending her entire income:

$$P_C C + P_M M = Y.$$

- Suppose a Kaffeehaus coffee costs €4, a cinema ticket costs €8, and Lena's monthly "fun budget" is €96.
- Maximum coffees (0 cinema): $96/4 = 24$. Maximum cinema visits (0 coffee): $96/8 = 12$.
- Slope of the budget line: $-P_M/P_C = -8/4 = -2$ — each extra cinema visit costs 2 coffees.

Constrained Choice: The Tangency Condition

- Lena's optimal bundle is where her highest reachable indifference curve is **tangent** to her budget line.
- At that point:
 $MRS_{C,M} = P_M/P_C$.
- With $U = \sqrt{C \times M}$ and this budget: optimum is $M = 6$, $C = 12$ (she splits her €96 evenly: €48 on each good).



Income and Substitution Effects

Suppose the price of cinema tickets rises. This has two distinct effects on demand:

Substitution effect

Holding utility constant, a relative price rise *always* causes less consumption of that good.

Income effect

A price rise makes the consumer effectively poorer, typically reducing consumption of *all* normal goods.

For a normal good, both effects push demand in the *same* direction — toward less of the good whose price rose.

Normal vs. Inferior Goods

Normal good

Demand *increases* with income (most goods).

Inferior good

Demand *decreases* with income.

Do you have an example of an inferior good?

2.2 Putting the Tools to Work: Sozialhilfe and Labor Supply

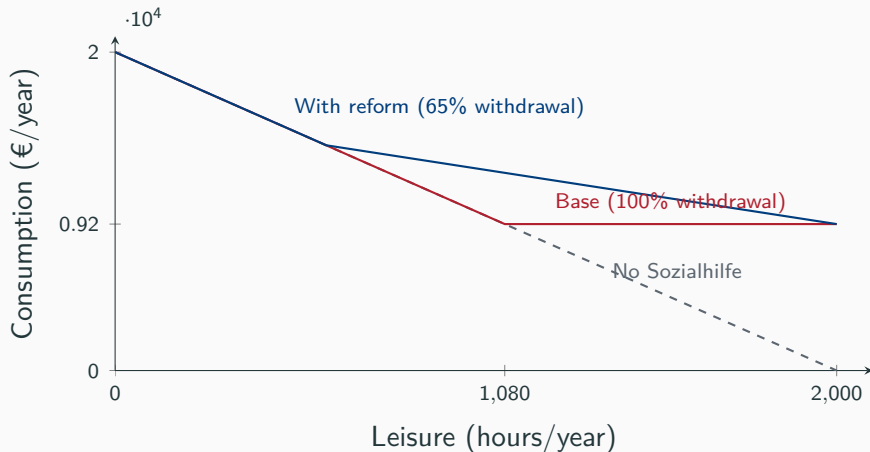
Austria's Sozialhilfe (Minimum Income Support)

- Since the 2019 *Sozialhilfe-Grundsatzgesetz*, Austria's means-tested minimum income sets **maximum** rates (*Höchstsätze*), not guaranteed minimums, with details varying by *Land*.
- Maximum monthly rate for a single adult/single parent (2024): $\approx$ €1,156. Average *actual* payment to single adults: $\approx$ €766/month.
- A 2019 reform introduced the *Freibetrag*: up to 35% of net earnings are disregarded (not counted against the benefit) for up to 12 months when a recipient starts working.

The Consumption–Leisure Trade-off

- Suppose Julia, a single parent, can work up to 2,000 hours/year at a wage of €10/hour, with no other income and no Sozialhilfe.
- Each hour of leisure “costs” her €10 in forgone wages (opportunity cost).
- Her budget constraint: maximum €20,000/year in consumption (0 leisure) or 2,000 hours of leisure (€0 consumption); slope = -10 .

Adding Sozialhilfe to the Budget Constraint



Without any earnings disregard, Julia's net return to work is €0 until she earns her way off Sozialhilfe entirely — a pure “welfare trap.”

The Wiedereinsteigerfreibetrag: A Real Reduction-Rate Experiment

- **Base regime:** 100% benefit-reduction rate. Every euro earned reduces Sozialhilfe by one euro — net return to work is €0 (flat budget segment).
- **Reform regime:** 35% of net earnings disregarded $\Rightarrow$ effective reduction rate $\approx 65\%$. Net return to work rises to €3.50 per hour (slope -3.5 instead of flat).

Predicted Effects on Labor Supply

- Cutting the reduction rate (100% $\rightarrow$ 65%) makes work pay more — a pure **substitution effect** toward less leisure (more work).
- But it also raises Julia's income at any given hours choice — a competing **income effect** toward *more* leisure.
- The net effect depends on the relative strength of the two — and on how much Julia values leisure relative to consumption (her preferences).

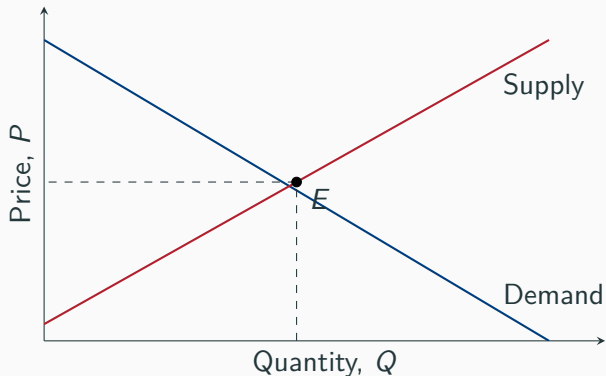
2.3 Equilibrium and Social Welfare

From Individual Choice to Market Demand

- Each individual's utility maximization generates her own demand $X(P, Y)$ for a good.
- The **market demand curve** is the horizontal sum of all individual demands at each price.
- **Elasticity of demand:** $\varepsilon_D = \frac{\% \Delta \text{quantity demanded}}{\% \Delta \text{price}}$ — typically negative, and typically not constant along the curve.

- Firms use a **production function** to turn inputs (labor, capital) into output, subject to **diminishing marginal productivity**.
- Profit maximization ($\max_Q [p \cdot Q - c(Q)]$) implies firms produce until **marginal cost equals price**.
- The marginal cost curve *is* the firm's supply curve; market supply is the horizontal sum across firms.

Market Equilibrium



Market equilibrium: the unique price/quantity pair at which demand equals supply, and both sides of the market are satisfied.

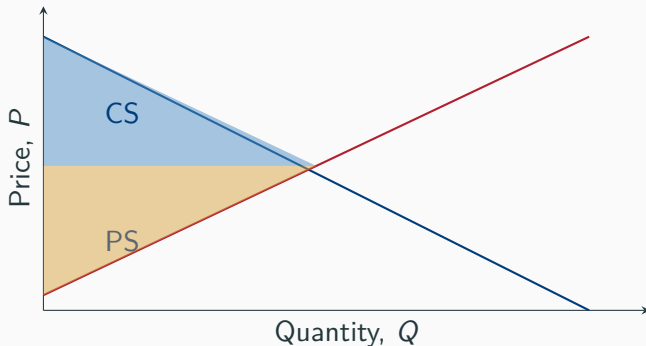
Consumer and Producer Surplus

Consumer surplus

The value consumers get from a good, above and beyond what they paid. Area below demand, above P_E .

Producer surplus

The profit producers earn, above and beyond their cost. Area above supply, below P_E .



The First Fundamental Theorem of Welfare Economics

First Fundamental Theorem

The competitive equilibrium, where supply equals demand, maximizes total economic efficiency.

1st Welfare Theorem holds if (1) no externalities, (2) perfect competition [individuals and firms are price takers], (3) perfect information, (4) agents are rational, then private market equilibrium is Pareto efficient

- Total surplus just counts € — it is blind to *who* gets them.
- When these assumptions fail (market failures), government intervention can potentially raise — not just redistribute — total surplus.

Application: Rent Regulation in Vienna

- About **21%** of Vienna's rented apartments are subject to the *Richtwertmietzins* price ceiling (currently €6.74/m²), below the market rent.
- If this price is below the equilibrium price, what happens to apartment supply?

Second Fundamental Theorem

Any Pareto-efficient allocation can be reached by (1) suitably redistributing initial endowments via **lump-sum** transfers, then (2) letting markets work freely.

If this worked in practice, there would be *no conflict* between efficiency and equity.

Why the Second Theorem Fails in Practice

- Lump-sum redistribution requires the government to observe individual *characteristics* (need, ability) — not just *outcomes* (income, work status).
- In reality, the government usually cannot tell these apart, and must base taxes/transfers on observable behavior instead.

Social Welfare Functions: Utilitarian

With a utilitarian social welfare function, society's goal is to maximize the sum of individual utilities:

$$SWF = U_1 + U_2 + \cdots + U_N$$

The utilities of all individuals are given equal weight, and summed to get total social welfare

If marginal utility of money decreases with income (satiation), utilitarian criterion values redistribution from rich to poor

Taking 1 for a rich person decreases his utility by a small amount, giving the 1 to a poor person increases his utility by a large amount Transfers from rich to poor increase total utility

Social Welfare Functions: Rawlsian

Rawls (1971) proposed that society's goal should be to maximize the well-being of its worst-off member. The Rawlsian SWF has the form:

$$SWF = \min(U_1, U_2, \dots, U_N)$$

Since social welfare is determined by the minimum utility in society, social welfare is maximized by maximizing the well-being of the worst-off person in society (=maxi-min) Rawlsian criterion is even more redistributive than utilitarian criterion: society wants to extract as much tax revenue as possible from the middle and rich to make transfers to the poor as large as possible

Other Social Justice Principles

1. **Commodity egalitarianism:** society should guarantee basic needs (education, health care, retirement) as rights; beyond that, distribution doesn't matter.
2. **Equality of opportunity:** individuals should be compensated for circumstances beyond their control (family background, ability) — but not for the consequences of their own choices.
3. **Just deserts:** compensation should match contribution — taxes should track the benefits an individual receives from government (a libertarian view).

None of these reduce cleanly to utilitarian or Rawlsian social welfare — real policy debates mix all three.

What Do People Actually Think Is Fair?

Saez & Stantcheva (2016) surveyed $\approx 1,100$ US respondents with hypothetical vignettes: two people have identical pre-tax and post-tax income — who deserves a €1,000 tax break more?

- People favor redistribution if they feel inequalities are “unfair” but views on what is fair differ
- Redistribution supported when people don’t have control [education for children, health insurance for the sick]
- Less support when people have some or full control [unemployment, being low income]
- Less support when people don’t “belong” (us vs. them)

Source: Saez, E. and Stantcheva, S. (2016), “Generalized Social Marginal Welfare Weights for Optimal Tax Theory,” *American Economic Review*.

Thank you!

Questions and discussion